Scenario-Based Questions | Set 4

1. A finance app wants to predict whether a loan will be approved...

Scenario: A finance app wants to predict whether a loan will be approved using data like income, credit score, employment type, and more. How can they choose only impactful features?.

Answer / Logic:

1. Collect income, credit score and employment details, and clean missing values.

2. Convert categorical details into suitable numerical features.

3. Use SelectKBest or Random Forest importance to find useful loan approval predictors.

4. Test the model with selected features and keep the ones that give reliable results.

2. A food delivery app wants to recommend meals to users based on the...

Scenario: A food delivery app wants to recommend meals to users based on the time of day and what they previously ordered. How should they implement this?

Answer / Logic:

1. Record what meals users order and whether they order at breakfast, lunch or dinner.

2. Build a user-meal history and identify their usual food choices.

3. Combine past-order recommendations with meals popular at that time of day.

4. Suggest suitable meals and update the list based on new orders.

3. A telecom company is predicting network faults based on technical...

Scenario: A telecom company is predicting network faults based on technical logs from 300+ parameters. How can they select the most relevant features?

Answer / Logic:

1. Clean the technical logs and handle missing or repeated readings.

2. Remove constant and highly similar parameters from the 300+ features.

3. Use SelectKBest or Random Forest feature importance to shortlist fault indicators.

4. Compare prediction results and keep the smaller set if accuracy stays good.

4. An online platform wants to recommend learning paths based on a...

Scenario: An online platform wants to recommend learning paths based on a student's current skills, goals, and previous course completions. How should this be implemented?

Answer / Logic:

1. Collect the student's current skills, goals and completed courses.

2. Add skill levels and topic tags to every available course.

3. Use content-based matching to find courses that fit the student's next step.

4. Recommend a learning path and update it after each course completion.

5. A health tracker app wants to predict sleep quality using 100+...

Scenario: A health tracker app wants to predict sleep quality using 100+ sensor features like heart rate, noise level, light exposure, etc. How should they reduce the feature space?

Answer / Logic:

1. Check sensor readings for missing values and unusual measurements.

2. Remove duplicate and low-variance features from the 100+ inputs.

3. Use feature selection to identify useful signals; try PCA if many are related.

4. Compare sleep prediction results before choosing the reduced feature set.

6. A travel booking website wants to recommend hotels based on...

Scenario: A travel booking website wants to recommend hotels based on ratings from users with similar travel styles. How should they design this?

Answer / Logic:

1. Collect hotel ratings and basic travel preferences from users.

2. Create a user-hotel rating matrix from their past stays.

3. Use collaborative filtering and cosine similarity to find users with similar tastes.

4. Recommend highly rated hotels they have not visited, filtered by budget and location.

7. A real estate company wants to estimate house prices using...

Scenario: A real estate company wants to estimate house prices using features like location, area, number of bedrooms, amenities, etc. How can they select important predictors?

Answer / Logic:

1. Prepare location, area, bedrooms and amenities for a house-price model.

2. Handle missing values and encode categories such as location.

3. Use f_regression with SelectKBest or Random Forest importance to rank features.

4. Compare prediction error and keep the most useful house-price predictors.

8. A music app wants to suggest songs based on user-selected mood...

Scenario: A music app wants to suggest songs based on user-selected mood (e.g., happy, calm, energetic). How can they build this recommendation?

Answer / Logic:

1. Add mood tags such as happy, calm and energetic to songs.

2. Record the mood selected by the listener.

3. Use content-based filtering to match songs with that mood.

4. Recommend matching songs and improve the list using listening history.

9. An HR department wants to predict employee attrition using...

Scenario: An HR department wants to predict employee attrition using demographic, performance, and feedback data. How can they reduce feature size 10.An e-commerce platform wants to recommend accessories (e.g., phone cases) based on the user's main product selection. How can this be implemented?

Answer / Logic:

1. Clean performance, feedback and other relevant employee data.

2. Remove duplicate or unnecessary columns and check data quality.

3. Use SelectKBest or a tree-based classifier to find useful attrition predictors.

4. Test the smaller feature set and check that the results are fair and reliable.

10. An e-commerce platform wants to recommend accessories (e.g., phone...

Scenario: An e-commerce platform wants to recommend accessories (e.g., phone cases) based on the user's main product selection. How can this be implemented?

Answer / Logic:

1. Record the main product's model, type and compatibility details.

2. Store matching accessory details, such as which phone cases fit each phone.

3. Use product attributes and frequently bought-together patterns to find suitable items.

4. Recommend compatible accessories and remove unavailable products.